

BLAKE FARMAN

CURRICULUM VITAE

MATHEMATICS AND STATISTICS
LOUISIANA TECH UNIVERSITY
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CONTACT INFORMATION

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Google Scholar: <https://scholar.google.com/citations?user=ySXYuuQAAAAJ>
arXiv: https://arxiv.org/a/farman_b_1

CITIZENSHIP

United States

RESEARCH INTERESTS

Algebra, (Noncommutative) Algebraic Geometry, Derived Categories, Category Theory, and Formalized Mathematics in Lean

EDUCATION

<i>Doctor of Philosophy</i> , Mathematics University of South Carolina, Columbia, SC Dissertation: Geometry of Derived Categories on Noncommutative Projective Schemes Advisor: Matthew Ballard	May 12, 2018
<i>Master of Science</i> , Mathematics University of Vermont, Burlington, VT Major Concentration: Abstract Algebra Minor Concentration: Algebraic Number Theory	May 22, 2011
<i>Bachelor of Science</i> , Computer Science Rensselaer Polytechnic Institute, Troy, NY Minors: Applied Mathematics and French	May 16, 2009

APPOINTMENTS

<i>Assistant Professor</i> Louisiana Tech University, Ruston, LA	September 2024 – Present
<i>Assistant Professor</i> University of Louisiana at Monroe, Monroe, LA	August 2020 – August 2024
<i>Capital One Endowed Professorship in Mathematics</i> University of Louisiana at Monroe, Monroe, LA	August 2020 – May 2023
<i>Visiting Assistant Professor</i> Lafayette College, Easton, PA	July 2018 – July 2020
<i>Graduate Teaching Assistant</i> University of South Carolina, Columbia, SC	August 2012 – May 2018
<i>Graduate Teaching Assistant</i> University of Vermont, Burlington, VT	August 2009 – May 2011

ADMINISTRATIVE APPOINTMENTS

Coordinator, M.S. in Mathematics September 2026 – Present
Mathematics and Statistics
Louisiana Tech University

LEADERSHIP AND GOVERNANCE

Communication Coordinator (Officer) February 2025 – Present
Louisiana-Mississippi Section
Mathematical Association of America

Membership Outreach Director (Officer) January 2025 – Present
Board of Directors
Friends of Black Bayou
Black Bayou National Wildlife Refuge
Monroe, LA

Director at Large January 2024 – January 2025
Board of Directors
Friends of Black Bayou
Black Bayou National Wildlife Refuge
Monroe, LA

EMPLOYMENT

Associate August 2011 – August 2012
Princeton Consultants, Inc, Princeton, NJ
Information Technology consultant.

PUBLICATIONS

Clifton, A. W., Farman, B., 2026. WIP: Using Career Panels to Spark Curiosity and Career Awareness Among Mathematics Majors, in: 2026 ASEE Annual Conference & Exposition Proceedings. Presented at the 2026 ASEE Annual Conference & Exposition, ASEE Conferences, Charlotte, North Carolina, p. 60813. <https://doi.org/10.18260/1-2--60813>.

Clifton, A. W., Fendley, M., Farman, B., Coriell, J., 2025. Work in Progress: First-Year Engineering Students' Confidence in Communicating Mathematical Content, in: 2025 ASEE Annual Conference & Exposition Proceedings. Presented at the 2025 ASEE Annual Conference & Exposition, ASEE Conferences, Montreal, Quebec, Canada, p. 57495. <https://doi.org/10.18260/1-2--57495>.

Farman, B., Clifton, A. W., Long, W. C., Corbett Cruse, K., 2025. Fostering Growth Mindsets: Implementing Standards-Based Grading in College Algebra, in: 2025 ASEE Annual Conference & Exposition Proceedings. Presented at the 2025 ASEE Annual Conference & Exposition, ASEE Conferences, Montreal, Quebec, Canada, p. 56592. <https://doi.org/10.18260/1-2--56592>.

Ballard, M., Farman, B., 2021. Kernels for noncommutative projective schemes. *J. Noncommut. Geom.* 15, 1129–1180. <https://doi.org/10.4171/jncg/427>.

Farman, B., 2018. Geometry of Derived Categories on Noncommutative Projective Schemes. Ph.D. dissertation. University of South Carolina. <https://scholarcommons.sc.edu/cgi/viewcontent.cgi?article=5742&context=etd>.

FORMALIZATION

Peer-reviewed contributions to the **Mathlib** library:

Introduced torsion theories for abelian categories. Work In Progress
CategoryTheory/Abelian/TorsionTheory; PR #36744.

Closure properties of orthogonal complements and S. E. Dickson's theorem. <code>CategoryTheory/ObjectProperty</code> ; PR #44226.	Work In Progress
Short exact sequences from pulling back a subobject. <code>CategoryTheory/Abelian</code> ; PR #44229.	Work In Progress
Kernels of pullbacks of subobjects. <code>CategoryTheory/Subobject</code> ; PR #44223.	Work In Progress
Introduced and characterized radicals on abelian categories. <code>CategoryTheory/Abelian/Preradical</code> ; Commit 09d467d	March 16, 2026
Added Stenström's colon construction for preradicals. <code>CategoryTheory/Abelian/Preradical</code> ; Commit c505a69	March 6, 2026
Introduced the basic theory of preradicals on abelian categories. <code>CategoryTheory/Abelian/Preradical</code> ; Commit 1f308dd	February 26, 2026
Defined the topologies associated to ideal filters. <code>RingTheory/IdealFilter</code> ; Commit a91e88c	February 2, 2026
Introduced ideal filters and Gabriel filters on the lattice of ideals of a ring. <code>RingTheory/IdealFilter</code> ; Commit 78484be	January 22, 2026
Generalized <code>Submodule.colon</code> to arbitrary subsets and upgraded the surrounding API. <code>RingTheory/Ideal</code> ; Commit 95093e9	January 16, 2026

SOFTWARE AND OPEN EDUCATIONAL RESOURCES

<i>dgcat-blueprint</i>	2026 – Present
Lean blueprint and dependency graph for the formalization of dg-categories and the derived category of a dg-category, produced under the Anthropic AI for Science award. https://github.com/LATechLean/dgcat-blueprint	
<i>ITPLxS25</i>	2025
Course materials for Interactive Theorem Proving with Lean (Math 490/587). https://github.com/LATechLean/ITPLxS25	

FUNDING

<i>Formalizing the dg-Enhancement of Triangulated Categories</i>	September 2026
Anthropic AI for Science	\$20,000
Six-month award of Claude usage credits to formalize dg-categories and the derived category of a dg-category in Lean's <code>Mathlib</code> library. Role: Principal Investigator (PI).	
<i>Applied eXploration In Open Mathematics Lab [AXIOM Lab]</i>	June 2025 – June 2026
Louisiana Board of Regents Support Fund (LEQSF(2025-26)-ENH-DE-07)	\$60,875
Targeted Departmental Enhancement Grant to establish a dedicated research collaboration lab within the Mathematics and Statistics program at Louisiana Tech University. Role: Principal Investigator (PI).	
<i>Student Activity Enhancement Fee</i>	February 2024
Student Development, University of Louisiana at Monroe	\$2,000
SAEF Grant to take mathematics students to the annual meeting of the Louisiana-Mississippi section of the Mathematical Association of America in New Orleans, LA.	
<i>Capital One Endowed Professorship in Mathematics (2022 - 2023)</i>	\$5,370
<i>Capital One Endowed Professorship in Mathematics (2021 - 2022)</i>	\$5,070
<i>Capital One Endowed Professorship in Mathematics (2020 - 2021)</i>	\$4,790

SUBMITTED GRANTS

<i>Formalizing the Recognition Theorem for Cofibrantly Generated Model Categories in Lean: Foundations for Homotopy Theory and Derived Algebraic Geometry (Declined)</i>	October 2025 \$69,110
Research Competitiveness Subprogram Louisiana Board of Regents	
<i>LEAPS-MPS: Building the Foundations of Formal (Declined)</i>	April 2025 \$165,753
<i>Homotopy Theory through Model Categories (FY 25)</i>	
National Science Foundation	
<i>Fourier-Mukai Partners in Noncommutative Algebraic Geometry (Declined)</i>	October 2024 \$118,624
DMS - Algebra and Number Theory National Science Foundation	
<i>Fourier-Mukai Partners in Noncommutative Algebraic Geometry (Declined)</i>	November 2023 \$113,918
Research Competitiveness Subprogram, FY 24 Louisiana Board of Regents Support Fund	
<i>Fourier-Mukai Partners in Noncommutative Algebraic Geometry (Declined)</i>	November 2022 \$142,041
Research Competitiveness Subprogram, FY 23 Louisiana Board of Regents Support Fund	
<i>Derived Categories in Noncommutative Projective Algebraic Geometry (Declined)</i>	\$210,000
Proposed NSF Mathematical Sciences Post Doctoral Research Fellowship to work with Ragnar Buchweitz (University of Toronto). Later revised to work with Michael Weems (University of Glasgow)	
<i>Dean's Dissertation Fellowship Nominee (Declined)</i>	\$25,000
Nominated by the Department of Mathematics at the University of South Carolina for a fellowship from the College of Arts and Sciences to spend a year in residence at the Institute of Advanced Study in Princeton, NJ.	

AWARDS AND HONORS

<i>Education Award</i>	2025
College of Engineering and Science Louisiana Tech University	
<i>Project NExT (New Experiences In Teaching) Fellow</i>	Red '22
The Mathematical Association of America	
<i>Rensselaer Medalist</i>	2004
Scholarship to Rensselaer Polytechnic Institute (\$60,000).	

GRADUATE MENTORING AND RESEARCH SUPERVISION

<i>Emma Foster</i>	2026 – Present
Ph.D. Student (Committee Chair), Computational Analysis and Modeling Louisiana Tech University	
<i>Joshua Bush</i>	2025–2026
Ph.D. Student (Committee Chair), Computational Analysis and Modeling Louisiana Tech University	
<i>Josiah Norman</i>	Graduating Spring 2026
M.S. Student (Committee Member) Louisiana Tech University	

UNDERGRADUATE MENTORING AND RESEARCH SUPERVISION

<i>Joshua Robinson</i> Louisiana Tech University Co-mentored with Ann Clifton on a Cayley graph research project, <i>Structural and Topological Properties of Cayley Graphs</i> , presented at the Louisiana Tech Undergraduate Research & Service-Learning Symposium and awarded second place in the Student Paper Competition at the LA/MS Section meeting of the Mathematical Association of America.	Spring 2025
<i>Riya Shrestha</i> Emerging Scholars Program, University of Louisiana at Monroe Knot Theory research project focused on identifying a family of knots for which there is a winning strategy in the Region Unknot Game.	Spring 2024
<i>The Entrepreneurship Pelican Cup</i> Immerse AR Served as faculty advisor for one of six finalists in the Pelican Cup Entrepreneurship business competition.	Spring 2023
<i>Sagar Bhetuwal</i> Emerging Scholars Program, University of Louisiana at Monroe Number Theory research project focused on providing a formal verification of the proof of the Prime Number Theorem using the Lean proof assistant.	Spring 2021

TEACHING INNOVATIONS

<i>Mathematical Art Installation</i> With Ann Clifton and Glen Whitney (Studio Infinity), guided students in building a compound of five tetrahedra from road-cone barriers and radio-antenna mounting hardware, exhibited alongside the senior design showcase. Recognized, together with the AXIOM Lab, by the 2025 COES Education Award.	Spring 2025
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TEACHING EXPERIENCE

LOUISIANA TECH UNIVERSITY

Course Number	Course Name	Term
Math 5843	Algebraic Geometry	S26
Math 587	Interactive Theorem Proving with Lean	S25
Math 5083	Intro to Abstract Algebra	F25
Math 4083	Intro to Abstract Algebra	F25
Math 490	Interactive Theorem Proving with Lean	S25
Math 4903	Algebraic Geometry	S26
Math 308	Introduction to Linear Algebra	W25
Math 243	Calculus III	F24

UNIVERSITY OF LOUISIANA AT MONROE

Course Number	Course Name	Section
Math 4007	History of Mathematics	44692 (F22)
Math 3086	Modern Algebra	44367 (F21), 63395 (S23)
Math 3080	Real Analysis	64863 (S22), 44174 (F23)
Math 3073	Numerical Methods	45407 (F23)
Math 3040	Introduction to Algebra and Analysis	64086 (S24)
Math 2040	Foundations of Mathematics	65041 (S23)

Course Number	Course Name	Section
Math 2032	Calculus III	44879 (F22), 44879 (F24)
Math 2002	Applied Linear Algebra	41001 (S21), 64042 (F21), 64042 (S23)
Math 1032	Calculus II	40999 (F24)
Math 1031	Calculus I	63789 (S24), 40997 (F24)
Math 1018	Contemporary Mathematics	40992, 40994 (F21), 60847, 62225 (S22)
Math 1011	College Algebra	42181, 43002, 43013 (F20), 62690, 62691 (S21), 63057, 63167 (S24)
Math 1007	Contemporary Mathematics	45282, 45288 (F22), 64979 (S23), 45282, 45288 (F23)

LAFAYETTE COLLEGE

Course Number	Course Name	Sections
Math 272	Linear Algebra with Applications	01 (S20)
Math 162	Calculus II	06, 07 (S19), 01, 03 (F19)
Math 161	Calculus I	01, 04 (F18), 04 (F19), 01, 02 (S20)
Math 161	Summer Program to Advance Leadership	SPAL (Su19)

UNIVERSITY OF SOUTH CAROLINA

Course Number	Course Name	Sections
Math 170	Finite Mathematics	007 (S14), 002 (S16)
Math 142	Calculus II	004 (Su14), 020 (S18)
Math 122	Calculus for Business Administration and Social Sciences	E01 (S17)
Math 116	Brief Precalculus Mathematics	E1B, E2B (F13), E1B, E2B (F15), E1B, E2B (F16)
Math 115	Precalculus Mathematics	011 (F14)
Math 111	Basic College Mathematics	M13 (Su13), 006 (S15), 006 (F17)

UNIVERSITY OF VERMONT

Course Number	Course Name	Sections
Math 019	Fundamentals of Calculus I	S, W(F10)
Math 017	Applications of Finite Math	G (F09), A (S10), Z2 (Su10)

PRESENTATIONS

<i>Formal Methods in Modern Algebra</i>	October 9, 2025
Computational Analysis and Modeling Seminar	
Louisiana Tech University, Ruston, LA	
<i>A Little Algebraic Topology</i>	April 11, 2025
Faculty Seminar	
COES Graduate Student Council	
Louisiana Tech University, Ruston, LA	
<i>Fostering Growth Mindsets: Implementing Standards-Based Grading in College Algebra</i>	February 22, 2025
Annual Meeting of the Louisiana/Mississippi Section	
Mathematical Association of America	
Jackson, MS	

<i>Algebra, Geometry, and Categories, Oh My!</i> Louisiana Tech University Ruston, LA	May 11, 2023
<i>Implementing Standards Based Grading in STEM</i> University of Louisiana System For our Future Conference 2023 Lake Charles, LA	March 28, 2023
<i>Algebra, Geometry, and Categories, Oh My!</i> Department Seminar, Lafayette College, Easton, PA	March 11, 2020
<i>A Little Abstract Nonsense</i> Lafayette College, Easton, PA	May 21, 2018
<i>Abstract Nonsense: Grokking Category Theory</i> Wilkes University, Wilkes-Barre, PA	May 17, 2018
<i>Kernels for Noncommutative Projective Schemes</i> Graduate Colloquium University of South Carolina, Columbia, SC	November 28, 2017
<i>Kernels for Noncommutative Projective Schemes</i> Algebra Seminar University of New Brunswick, Fredericton, NB, CA	October 12, 2017
<i>Kernels for Noncommutative Projective Schemes</i> Algebra Seminar Temple University, Philadelphia, PA	October 9, 2017
<i>Kernels for Noncommutative Projective Schemes</i> Algebraic Geometry Seminar University of South Carolina, Columbia, SC	August 28, 2017
<i>Noncommutative Projective Schemes</i> Temple Graduate Student Conference in Algebra, Geometry, and Topology Temple University, Philadelphia, PA	May 13, 2016
<i>Noncommutative Projective Schemes</i> Student Algebra/Number Theory Seminar University of South Carolina, Columbia, SC	May 14, 2016
<i>Fully Faithful Abstract Nonsense:</i> <i>Representable Functors and The Yoneda Embedding</i> Graduate Colloquium University of South Carolina, Columbia, SC	November 2015
<i>Fully Faithful Abstract Nonsense:</i> <i>Representable Functors and The Yoneda Embedding</i> Graduate Student Algebraic Geometry, Commutative Algebra, and Number Theory Seminar University of South Carolina, Columbia, SC	July 2015

CONFERENCES AND WORKSHOPS

<i>LA/MS Sectional Meeting</i> Mathematical Association of America Jackson, MS	February 20 – February 22, 2025
<i>Math Corequisite Academy Workshop</i> Louisiana Board of Regents Baton Rouge, LA	October 15, 2024
<i>LA/MS Sectional Meeting</i> Mathematical Association of America New Orleans, LA	February 29 – March 2, 2024

<i>High School Math Teacher and Faculty Collaboration</i> Louisiana Department of Education Baton Rouge, LA	November 30, 2023
<i>Math Corequisite Academy Workshop</i> Louisiana Board of Regents Baton Rouge, LA	November 7, 2023
<i>Mathematical Association of America MathFest</i> Tampa, FL	August 1–6, 2023
<i>Mathematics Research Community</i> Derived Categories, Arithmetic and Geometry, Bridgeland Stability Group Beaver Hollow Conference Center, Java, NY	June 4–10, 2023
<i>Joint Mathematics Meetings</i> Boston, MA	January 3–7, 2023
<i>Mathematical Association of America MathFest</i> Philadelphia, PA	August 3–6, 2022
<i>Mathematical Association of America Project NExT (New Experiences in Teaching)</i> Philadelphia, PA	August 1–3, 2022
<i>Series on Exploring Combinatorics and Number Theory (SECANT) IV</i> Cedar Crest College, Allentown, PA (Online)	January 15–16, 2022
<i>Derived, Birational, and Categorical Algebraic Geometry</i> Invited Participant Banff International Research Station, Banff, Alberta, CA	November 14–19, 2021
<i>Series on Exploring Combinatorics and Number Theory (SECANT) III</i> Cedar Crest College, Allentown, PA (Online)	January 15–16, 2021
<i>Derived, Birational, and Categorical Algebraic Geometry</i> Invited Participant Banff International Research Station, Banff, Alberta, CA (Online)	November 2–6, 2020
<i>Joint Mathematics Meetings</i> Denver, CO	January 15–17, 2020
<i>Series on Exploring Combinatorics and Number Theory (SECANT) II</i> Cedar Crest College, Allentown, PA	October 12–13, 2019
<i>Joint Mathematics Meetings</i> Baltimore, MD	January 16–19, 2019
<i>Series on Exploring Combinatorics and Number Theory (SECANT)</i> Cedar Crest College, Allentown, PA	November 17–18, 2018
<i>Georgia Algebraic Geometry Symposium</i> Georgia Institute of Technology, Atlanta, GA	February 23–25, 2018
<i>Workshop on Topics in Algebraic Geometry</i> The University of North Carolina at Chapel Hill, Chapel Hill, NC	November 3–5, 2017
<i>Palmetto Number Theory Series (PANTS) XXVIII</i> University of Tennessee, Knoxville, TN	September 16–17, 2017
<i>Georgia Algebraic Geometry Symposium</i> University of Georgia, Athens, GA	March 3–5, 2017
<i>Workshop on Symplectic Varieties and Geometric Representation Theory</i> The University of North Carolina at Chapel Hill, Chapel Hill, NC	October 28–30, 2016
<i>UGA-SWAG: Summer Workshop in Algebraic Geometry</i> University of Georgia, Athens, GA	August 26–29, 2016
<i>Temple Graduate Student Conference in Algebra, Geometry, and Topology</i> Temple University, Philadelphia, PA	May 14–15, 2016

<i>Georgia Algebraic Geometry Symposium</i> Emory University, Atlanta, GA	October 2015
<i>AMS Summer Institute on Algebraic Geometry</i> University of Utah, Salt Lake City, UT	July 27–31, 2015
<i>Georgia Algebraic Geometry Symposium</i> University of Georgia, Athens, GA	October 2014
<i>Arizona Winter School</i> University of Arizona, Tucson, AZ	March 2014
<i>Palmetto Number Theory Series (PANTS) XXI</i> Clemson University, Clemson, SC	December 2013
<i>Palmetto Number Theory Series (PANTS) XX</i> Davidson College, Davidson, NC	September 2013
<i>Algebraic Geometry Northeastern Series (AGNES)</i> Yale University, New Haven, CT	April 2013
<i>AMS Spring Western Sectional Meeting</i> University of Colorado, Boulder, CO	April 2013
<i>Palmetto Number Theory Series (PANTS) XIX</i> University of South Carolina, Columbia, SC	December 2012
<i>Palmetto Number Theory Series (PANTS) XVIII</i> Wake Forest University, Winston-Salem, NC	September 2012

PROFESSIONAL MEMBERSHIP

<i>Mathematical Association of America</i>	Member
<i>American Mathematical Society</i>	Member

SERVICE

TO THE PROFESSION

<i>Section NExT Committee Member</i> Louisiana-Mississippi Section Mathematical Association of America	February 2025 – Present
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TO THE STATE

<i>Mindset Meauxtivorator</i> Louisiana Board of Regents	Fall 2024 – Spring 2025
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TO THE UNIVERSITY

LOUISIANA TECH UNIVERSITY

<i>Faculty Senate</i> College of Engineering and Science Representative; member of the subcommittee revising Policy 1301	2025–2026
<i>Undergraduate Advising</i> Advisees: J. D. Vessel and Bryce Labbie	2025–2026
<i>Career Panels for Mathematics Majors</i> Collaborated with Ann Clifton and the Mathematics Club to organize career panels for mathematics majors.	2025–2026
<i>Literary Rally</i> Proctor	March 15, 2025

<i>Undergraduate Research & Service-Learning Symposium</i> Judge	March 14, 2025
<i>Professors on the Plaza</i>	October 16, 2024
UNIVERSITY OF LOUISIANA AT MONROE	
<i>Browse on the Bayou</i> Mathematics program representative for ULM's Fall on-campus recruiting event	September 30, 2022
<i>Mathematics Hiring Committee Chair</i>	Fall 2023
<i>Mathematics Hiring Committee Member</i>	Fall 2022 – Spring 2023
<i>Co-Requisite Course Design</i> Contemporary Mathematics	Spring 2022
<i>Browse on the Bayou</i> Mathematics program representative for ULM's Spring on-campus recruiting event	March 21, 2022
<i>Investiture Ceremony Marshal</i>	October 1, 2021
<i>University Curriculum Committee Member</i>	2021–2024
<i>Louisiana Math Forward Committee Member</i>	2021
<i>Mathematics Hiring Committee Member</i>	Spring 2021
UNIVERSITY OF SOUTH CAROLINA	
<i>High School Math Contest Proctor</i>	2013–2015
TO THE COMMUNITY	
<i>National Hunting and Fishing Day</i> Represented Friends of Black Bayou at the Louisiana Department of Wildlife and Fisheries event for National Hunting and Fishing day. Black Bayou Lake National Wildlife Refuge	September 27, 2025
<i>Fall Celebration</i> Setup, Operated Canoe/Kayak Rental, Breakdown Black Bayou Lake National Wildlife Refuge	October 19, 2024
<i>Water Sweep</i> Waterway cleanup at Black Bayou Lake National Wildlife Refuge Ouachita Green	September 14, 2024
ORGANIZER	
<i>Student Algebra/Number Theory Seminar</i> University of South Carolina	Summer 2016
<i>Student Algebraic Geometry, Commutative Algebra, and Number Theory Seminar</i> University of South Carolina	Summer 2015
<i>Student Algebraic Geometry Seminar</i> University of South Carolina	Spring 2015
<i>Student Algebraic Geometry Seminar</i> University of South Carolina	Fall 2014